

Omar Alterkait Resume

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OmarAlterkait.github.io

OBJECTIVE	<i>PhD candidate developing foundation models for sparse 3D data and differentiable programming frameworks. Seeking ML research internship in interpretable ML, computer vision, or ML for science/healthcare, with expertise in inverse problems, multi-modal sensor fusion, geometric deep learning, and self-supervised learning at scale.</i>	
EDUCATION	Tufts University PhD Physics MS Physics	Expected May, 2028 February, 2023
	University of Colorado, Boulder BA Physics (Magna Cum Laude) BS Applied Mathematics	May, 2021
TECHNICAL SKILLS	Python (NumPy, JAX, Matplotlib, Pytorch, HDF5, Plotly, Pandas, Uproot, ...), LaTeX, Containers, Git/Github, Mathematica, C++, ROOT, Linux, C, MATLAB, Excel (VBA), Arabic Calligraphy, and Fluent in both English and Arabic.	
ML EXPERTISE	Methods: Differentiable Programming, Sparse 3D Models (MinkowskiNet, PTV3), Equivariant Neural Networks (E3NN), Foundation Models, Self Supervised Learning, Hyperparameter Tuning Scale: 10M training samples, Multi-GPU JAX/PyTorch, 100 – 10,000× speedups Applications: 3D Inverse Problems, Multi-Modal Sensor Fusion, Differentiable Physics Simulation for Calibration and Reconstruction	
PREPRINTS	Omar Alterkait, Csar Jess-Valls, Ryo Matsumoto, Patrick de Perio, and Kazuhiro Terao. <i>End-to-end Differentiable Calibration and Reconstruction for Optical Particle Detectors</i> . arXiv:2602.24129 [hep-ex], 2026.	
WORKSHOP PAPERS	Omar Alterkait et al. <i>10 Million Particle Events: Enabling Foundation Models for Sparse 3D Inverse Problems</i> . NeurIPS 2025 AI for Science Workshop, Spotlight Poster.	
	Omar Alterkait and César Jesús Valls. <i>Differentiable Ray-Tracing for Optical Particle Detector Simulation</i> . NeurIPS 2025 Machine Learning and the Physical Sciences Workshop, Poster.	
POSITIONS	SCGSR-IRCA Fellow, CERN	July 2026
	SCGSR Fellow, SLAC	January-December 2026
	IAIFI Junior Investigator, MIT	August 2021-Present
	Ozaki Program Fellow, Kavli IPMU, UTokyo	August-December 2024
	Academic Scholar, Kuwait University	September 2023-Present
TALKS	IRIS-HEP Fellow	January-June 2021
	Neutrino Physics and Machine Learning	2023, 2024, 2025, 2026
	Nagoya University	November 2025
	Kyoto University	November 2025
	Tohoku University	October 2025
	Colorado State University	October 2025
	Fifth MODE Workshop on Differentiable Programming	June 2025
	APS Global Physics Summit	March 2025
POSTERS	Center for Data-Driven Discovery Kavli IPMU, UTokyo	March 2024
	Neutrino Physics and Machine Learning (Best Poster)	June 2026
	NEUTRINO	June 2026
	Neutrino Physics and Machine Learning (Best Poster)	November 2025
	IAIFI Summer Workshop	August 2025
	Fifth MODE Workshop on Differentiable Programming (Best Poster)	June 2025

EXPERIENCE

August 2021 - Present

- **Multi-Modal Foundation Models for 3D Reconstruction in LArTPCs** (w/ Kazuhiro Terao, SLAC) [github.com/OmarAlterkait/JAXTPC]
 - Generating first 10M event dataset with raw sensor waveforms paired with 3D ground truth ($30\times$ larger than existing)
 - Achieved $100\times$ speedup (50s to 0.5s per event) via JAX/GPU rewrite; captures charge+light dual-modality from same physics events
 - Processing extreme global sparsity with locally dense information; reconstructing 3D trajectories from 2D tomographic projections
 - Developing PTv3/Sonata transformers for efficient point cloud processing at scale; implementing cross-modal masked autoencoding directly on raw data
 - Releasing JAX framework with Data Olympics challenges to drive collaborative algorithm development and establish benchmarks
 - Planning mechanistic interpretability analysis to understand learned physics representations and inductive biases
- **Differentiable Ray-Tracing for Optical Detector Calibration (LUCiD)** (w/ César Jesús Valls, CERN) [github.com/CIDeR-ML/LUCiD]
 - Developed first differentiable ray-tracing framework achieving end-to-end differentiability from particle parameters to detector response
 - Models photon emission using neural representations trained on simulation data; handles stochastic interactions while maintaining differentiability
 - Achieved $10,000\times$ speedup over CPU Monte Carlo (1M photons in 30ms) via JAX automatic differentiation, JIT compilation, and grid-based spatial filtering
 - Matches state-of-the-art reconstruction baselines while enabling direct optimization through complex parameter landscapes where MCMC fails
 - Creating open-source benchmark dataset for ML-based detector reconstruction algorithm development
- **Sparse Euclidean Equivariant Neural Networks for Particle Reconstruction** (w/ Tess Smidt, MIT)
 - Developing sparse equivariant neural networks that preserve Euclidean symmetry (rotation, translation, reflection) eliminating need for data augmentation
 - Addressing extreme sparsity of LArTPC data by combining sparse convolutions with equivariant architectures
 - Targeting MicroBooNE detector particle reconstruction with improved sample efficiency through geometric inductive biases

PI : Kevin Stenson

- Addressed non-linear complexity scaling from increased collision rates using neural networks for real/fake track classification
- Processed high-dimensional detector data with extreme class imbalance in offline tracking pipeline